

LITERATURE REVIEW ON AUTONOMOUS UNDERWATER VEHICLE (AUV) SIMULATION USING GAZEBO

**A Project Report Submitted to the Defence Research and
Development Organisation (DRDO)**



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The objective was to understand how modern simulation framework bridges the gap between theoretical algorithms and physical deep sea deployment

What Are AUV?

AUVs are unscrewed ,self propelled underwater robots that operate without a physical tether to complete tasks without requiring real time control from a human

Applications

- 1.Defence:Underwater mines and surveillance
- 2.Scientific :Seafloor mapping and researching under polar ice shelves
- 3.Commercial: pipeline inspections

Real world sea trials	Gazebo simulation
Slow ,bounded by battery life and logistics	Instantaneous , scriptable and scannable
Dangerous ;risking the loss of a highly expensive robot	Safe ;collision simply reset the environment



WHY GAZEBO SIMULATION

- 1.simulate exactly how water resistance, buoyancy, and ocean currents will push and pull on the robot
- 2.Realistic Sensor Noise:adding realistic "noise" to train data, ensuring that computer vision and sensor algorithms learn to handle imperfect, real-world conditions
- 3.Seamless Code Deployment :developers don't have to change their code when moving from simulation to the real world

Gazebo Architecture

Gazebo has a decoupled (separated) architecture. The gzserver (brain) handles computational backend, generates sensors data and updates about robot's state. gzclient runs the graphical frontend. Draw 3D world on user interface so we can see the simulation on the screen

This difference is helpful as we don't have to run them together so while testing we can save computer processing power

uuv_simulator package uses gazebo plugins to calculate hydrodynamic forces. It treats water as external force that pushes against the different parts of robot

freefloating_gazebo package

Computes buoyancy and fluid resistance directly from the submerged volume, making it effective to exhibit irregular surface geometries

We want to train it for real world which has imperfect data thus we use plugins
We use plugins like DAVE to model aquatic sensors

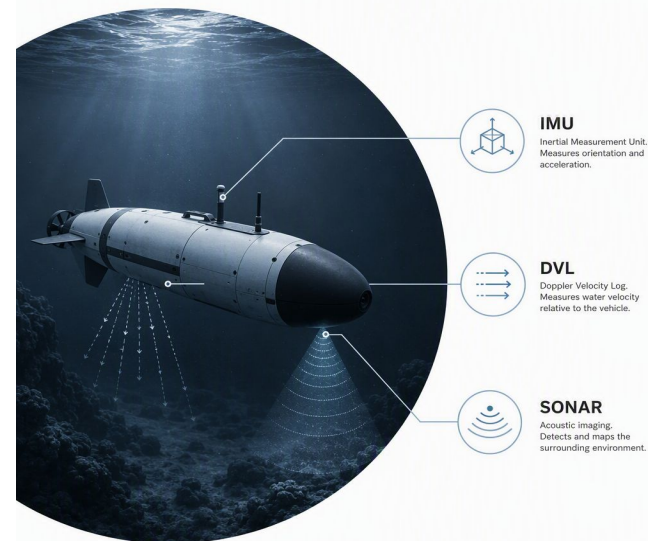
The Publish/Subscribe Architecture

1. Gazebo and ROS talk to each other using publish /subscribe system. Simulator sensors generate data and publish to ROS topics
2. AUV control software 'subscribes' to these topics, process info and publishes commands to actuator topics and move vehicle forward

The Transition to ROS2

Marine robotics field is migrating from ROS1 to ROS2 which uses a decentralised Data Distribution Service communication layer
Modern simulator use Project Plankton the official open source port for uuv_simulator built specifically for ROS2

It helps in real time processing, optimises multi robot network topologies which are crucial for Simulating synchronized AUV swarms



WHAT IS ROS?

It is an open-source **middleware framework** that runs *on top of* a traditional operating system. ROS provides a standard set of tools, libraries, and communication protocols that let different parts of a robot (called "nodes") talk to each other seamlessly.

AUV Dynamic Modelling

Coordinate Systems: Navigation relies on translating between two frames:
Earth-Fixed Frame: The static global map (used for waypoints and tracking drift).

Body-Fixed Frame: The dynamic map attached to the AUV's center of gravity (used for sensor readings and thruster commands).

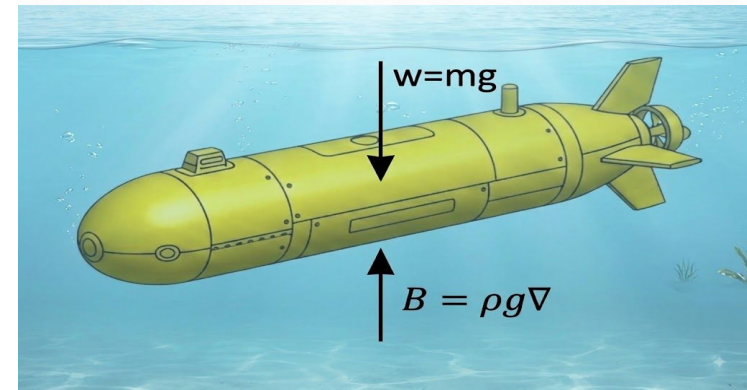
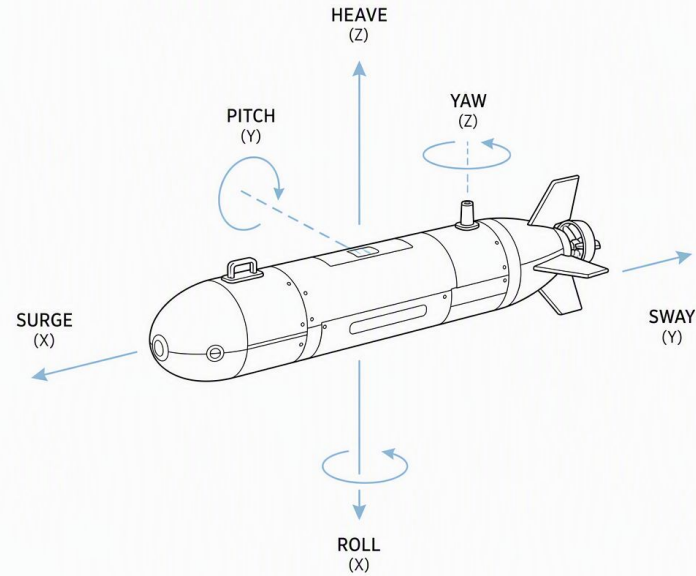
Hydrodynamic Modelling in Gazebo

Added Mass Matrix: Accelerating underwater requires pushing heavy water out of the way. Gazebo mathematically adds this "water weight" to the AUV's mass. A 6x6 mass is added to the dry weight to have the final weight

Hydrodynamic Damping: Simulates energy loss and drag (water resistance). Uses linear damping for slow speeds and quadratic drag for high-speed maneuvers, accounting for ocean currents.

Restoring Forces (Passive Stability): Simulates the tug-of-war between Gravity (pulling down) and Buoyancy (pushing up).

The Pendulum Effect: By placing the Center of Buoyancy above the Center of Gravity, the AUV acts like a pendulum, automatically righting itself if knocked off balance, without needing battery power.



Literature Review

In early stages research relied on physical testbeds. Those kinetic models cant capture messy marine environment

Gazebo breakthrough:it helped developers successfully bring advanced physics engines into underwater domain We have moved from ROS1 to ROS2 and have started using packages like freefloating_gazebo for real time processing. Via uuv_simulator package Gazebo's standard terrestrial physics can be extended using C++ plugins

As the foundation of AUV is trained now AI is being used to train models and it needs millions of trial and run errors To accurately test swarms simulation must restrict communication between agents to replicate the ocean .

Research Gaps

- 1.Modelling complex fluid dynamics to handle boundary layer interaction and non uniform turbidity
- 2.More better computation required for high speed and parallel execution required by deep learning and swarm development



Summary of Simulation Benefits

- **Cost and Risk Reduction:** Simulation drastically cuts the high costs, logistical challenges, and risks associated with physical sea trials.
- **Zero-Risk Testing:** Provides a safe virtual environment to test complex algorithms, preventing the loss of expensive physical prototypes.
- **Hardware-in-the-Loop (HIL):** Gazebo's modular architecture integrates seamlessly with frameworks like ROS and Rock, allowing developers to test actual vehicle hardware alongside virtual software.
- **System Validation:** Empowers engineers to fully validate Guidance, Navigation, and Control (GNC) systems using accurate 6-DOF physics, hydrodynamics, and simulated sensors before the robot ever touches the water.

Future Scope

- **Photorealistic Visuals:** Future simulators need native, out-of-the-box rendering for tricky optical effects (like water murkiness and light scattering) as AUVs rely more on cameras.
- **Swarm Coordination:** A critical need for better modeling of harsh underwater communication (high signal loss and low bandwidth) to test teams of coordinating robots.
- **Biomimetic and Soft Robotics:** Moving beyond traditional rigid-body physics to simulate flexible, animal-inspired designs, such as soft actuators and fin propulsion.
- **Advanced AI Training:** Simulators must be optimized to handle the massive processing loads required to train deep learning models and complex sensor fusion in unstructured environments.